

Database Management System

TABLE JOIN in SQL

A JOIN is used to combine rows from two or more tables based on a related column between them. Used for linking Students table with Departments table using dept_id.

Example Tables

Table 1: Students

student_id	name	dept_id
1	Rahim	101
2	Karim	102
3	Fahim	101
4	Abir	NULL

Table 2: Departments

dept_id	dept_name
101	Computer Science
102	Electrical Engg.
103	Mechanical Engg.

Types of JOIN

1. INNER JOIN: Returns only the rows where there is a match in both tables.

Syntax:

```
SELECT Students.name, Departments.dept_name
FROM Students
INNER JOIN Departments
ON Students.dept_id = Departments.dept_id;
```

Result:

name	dept_name
Rahim	Computer Science
Karim	Electrical Engg.
Fahim	Computer Science

2. LEFT JOIN (LEFT OUTER JOIN): Returns all records from the left table and matched records from the right table.

If no match, returns NULL.

Syntax:

```
SELECT Students.name, Departments.dept_name
FROM Students
LEFT JOIN Departments
ON Students.dept_id = Departments.dept_id;
```

Result:

name	dept_name
Rahim	Computer Science
Karim	Electrical Engg.
Fahim	Computer Science
Abir	NULL

3. RIGHT JOIN (RIGHT OUTER JOIN): Returns all records from the right table and matched records from the left table.

Syntax:

```
SELECT Students.name, Departments.dept_name
FROM Students
RIGHT JOIN Departments
ON Students.dept_id = Departments.dept_id;
```

Result:

name	dept_name
Rahim	Computer Science
Karim	Electrical Engg.
Fahim	Computer Science
NULL	Mechanical Engg.

4. FULL OUTER JOIN: Returns all records when there is a match in either table. If there is no match, fills with NULL. MySQL doesn't support FULL OUTER JOIN directly, in that case we will use UNION.

Syntax:

```
SELECT Students.name, Departments.dept_name
FROM Students
FULL OUTER JOIN Departments
ON Students.dept_id = Departments.dept_id;
```

5. CROSS JOIN: Returns the Cartesian product of both tables (all combinations).

Syntax:

```
SELECT Students.name, Departments.dept_name
FROM Students
CROSS JOIN Departments;
```

Result Example (Partial):

name	dept_name
Rahim	Computer Science
Rahim	Electrical Engg.
Rahim	Mechanical Engg.

6. SELF JOIN: A table joined with itself.

Example table: Employees

emp_id	name	manager_id
1	A	NULL
2	B	1
3	C	1

Syntax:

```
SELECT e1.name AS Employee, e2.name AS Manager
FROM Employees e1
LEFT JOIN Employees e2
ON e1.manager_id = e2.emp_id;
```